POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a Just-In-Time (JIT) compiler? Explain the architecture of Java and describe the roles of JDK, JRE, and JVM in the execution of a Java program. 7

   b) Define error and exception. Write a program to ask the user to enter two integers and perform division. Catch all relevant exceptions. 8

   OR

   Define collection framework. How do you create dynamic and unique list using collection framework with code example?

2. a) What is polymorphism? Write down the code to achieve compile time and runtime polymorphism in java using method overloading and overriding. 7

   b) Define an abstract class in Java. Explain the use of extends and super keyword with necessary example code. 8

3. a) Define Layout Management. Compare the different types of Swing and JavaFX layout in brief. 7

   b) Differentiate between AWT and Swing. Create suitable GUI using java swing to input first name and last name separately and finally display the full name when user clicks the submit button. 8

4. a) Create a TCP client server application where the client sends a string and the server responds by replying the length of the string. 7

   b) Define RMI stubs and skeleton. Create an RMI application where a client can remotely invoke a method that checks if the number is even or odd. 8

5. a) Explain the different types of JDBC drivers with diagrams. Discuss about SQL Injection and how PreparedStatement in Java overcomes this problem. 7

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b) Define JDBC. Write down the code to authenticate a user from database. 8

6. a) Define JSP and Servlet. Write down the necessary HTML and servlet code to display a username field and a button Greet. When user clicks on Greet button display "Welcome <<username>>" message in the web page. 7

b) What is ORM and how does it differ from JDBC? Also, define a web framework with examples and discuss the Spring framework in detail. 8

OR

What is private constructor? Explain the significance of private constructor in singleton design pattern with necessary example code.

7. Write short notes on: **(Any two)** 2×5

a) Multithreading in Java

b) Lifecycle of servlet

c) Transaction

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